// SYSTEM_LOG: NORTHBANK_DAY_2_OPS

Incident Report: Lab 04

SEV-2 Diagnostics: When instances run but applications fail.

Target Environment:	AWS Day 2 Operations
Module:	04
Clearance:	Cloud Engineers, DevOps, SREs

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The Anomaly: Total application failure on a healthy server.

SEV-2

Reporter: Marco (App Team)

Symptom: A web server shows healthy in EC2 but isn't serving pages.

Constraint 1: The AWS console claims the instance is perfectly healthy.

Constraint 2: The application is totally unresponsive.

⚠ Constraint 3 (Critical): There is no SSH key available to access the server.

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‘Running’ is not ‘Healthy’: Mapping the diagnostic domains.

System Status Check
(AWS Domain)



Owner: AWS

Meaning: Power is on, network is reachable, physical host is operational.

Instance Status Check
(OS Domain)



Owner: The Operating System

Meaning: The OS booted successfully, memory/CPU are accessible.

Target Health
(Application Domain)



Owner: The Application (e.g., Nginx)

Meaning: The actual service is responding to traffic.

The Mismatch: EC2 only reports on the System and OS. An application can completely fail while EC2 continues to report 'Running' and 'OK'.

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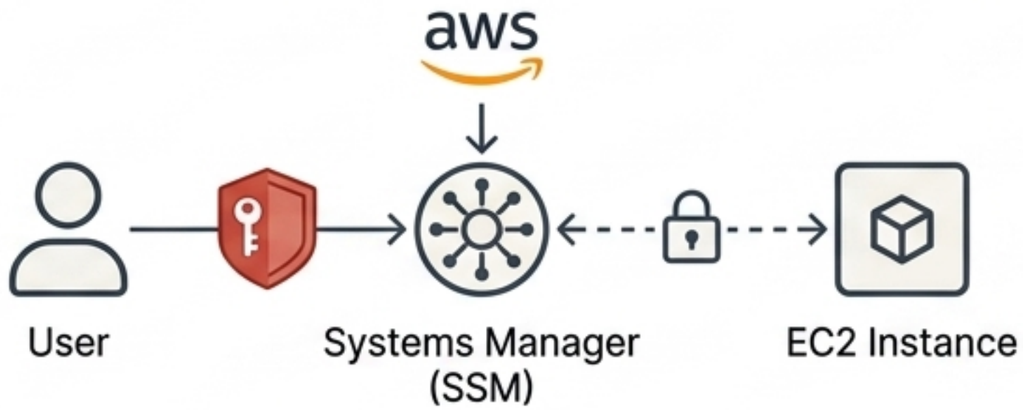
The Access Dilemma:
Why the missing SSH key is a feature, not a bug.

The Old Way



- Port 22 must remain openly exposed
- Relies on distributed static SSH keys
- High risk of credential leakage and lateral movement

The Modern Way



- No open inbound ports required
- No SSH keys exist to be leaked
- Fully logged, audited, and IAM-controlled access

Takeaway: Get in with SSM Session Manager — no keys, no open port 22.

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The Investigation: Verifying the telemetry mismatch in the terminal.

```
$ export ASG_NAME from terraform output
```

Intent: Target the Auto Scaling Group.

```
$ ./break.sh
```

Intent: Simulates the SEV-2 by stopping Nginx on the servers via SSM (verifying we need no SSH key to interact).

```
$ describe-instance-status
[OK] System Status  [OK] Instance Status

$ describe-target-health
[UNHEALTHY] Target Health
```

The anomaly **confirmed**: Infrastructure is blind to the application failure.

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The Execution: Bypassing the perimeter and restoring the service.

```
$ ssm start-session
```

Intent: Establish a secure shell without SSH keys.

```
$ systemctl status nginx
  Loaded: masked (Reason: Unit
nginx.service is masked.)
  Active: inactive (dead)
```

Intent: Inspect application state. The service is explicitly masked and stopped by the break script.

```
$ systemctl unmask + enable --now nginx
```

Intent: Force the OS to unmask, enable, and immediately start the web server in a single command.

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The Day-2 Dilemma: Do we fix the server, or destroy it?

Unhealthy instance identified in an Auto Scaling Group.

Tactical Path: Fix in Place via SSM

- **Path details:** Use systemctl to manually restore the service.
- **When to use:** Investigating root cause, debugging stateful applications, or preventing immediate system-wide failure cascades.

Strategic Path: Let the ASG replace it

- **Path details:** Terminate the EC2 instance; Auto Scaling Group provisions a fresh, healthy replacement.
- **When to use:** Stateless infrastructure ('cattle, not pets'), immutable architecture patterns.

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Verification: Infrastructure and Application are synchronized.

Telemetry Check: Target Health

Healthy

Telemetry Check: Web Endpoint

HTTP 200 OK

Ticket Closed: Nginx is successfully serving pages. The mismatch has been resolved.

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Day-2 Post-Mortem: Core principles for NorthBank operations



Visibility Principle: Running ≠ Healthy

Never rely solely on EC2 instance state. Always configure and monitor Target Health to ensure the application is actually serving traffic.



Security Principle: SSM Replaces SSH

Eliminate static keys and open ports. Session Manager provides logged, audited, and IAM-controlled access by default.

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